SAWVIAL

Technical Implementation and Preliminary Evaluation

Dataset engineering | Retrieval algorithms | Conversational RAG | Evaluation

PROJECT DOSSIER

A presentation-ready technical account of the SAWVIAL agricultural research assistant.

Prepared from the current source code, canonical dataset metadata, index metadata, and the saved seven-case live-API evaluation report.

186
research-paper records

15,764
knowledge units

6,563
dataset Q&A pairs

15,752
indexed vectors

Scope note

Preliminary engineering evaluation, not a scientific or agronomic validation study. Metrics are reported with benchmark limitations.

01 / PROJECT OVERVIEW

Executive summary

What has been implemented

SAWVIAL is a local, dataset-grounded agricultural research assistant. It turns research PDFs into traceable knowledge records; ranks evidence for a question; supplies selected records and recent dialogue to an instruction-tuned language model; and returns an answer with source metadata. It targets soil management, water management, and soil-water conservation rather than unrestricted general-purpose advice.

- A document-to-corpus pipeline with PDF/table parsing, sentence-aware chunking, metadata and context extraction, classification, numeric facts, provenance, quality states, and deterministic IDs.
- An offline lexical path based on normalized TF-IDF with unigram/bigram features and exact sparse similarity ranking; a FAISS artifact and explicit ID map are also persisted.
- A local Qwen2.5-1.5B-Instruct answer model by default; the LoRA adapter is opt-in. NLLB-200 translation is configured when enabled.
- A FastAPI service and browser chat UI with bounded history, follow-up handling, crop-specific evidence filtering, out-of-domain/no-evidence responses, cancellation, and speech controls.
- A live evaluation script reporting retrieval ranking, API grounding flag, citation ratio, format compliance, and latency.

Saved seven-case evaluation snapshot

Measure	Saved value	Interpretation
Live smoke cases	7	Small, targeted regression set.
Recall@5 / MRR / nDCG@5	1.0000 / 1.0000 / 1.0000	Gold IDs appeared first and all expected IDs were retrieved in these cases.
Grounded flag rate	1.0000	API field, not a factuality judge.
Cited-source ratio	0.3143	Citation coverage is incomplete.
One-line compliance	1.0000	Both one-line cases passed.
Mean latency	26,730.8 ms	About 26.7 s per request in this saved run.

Interpretation

Top-rank retrieval on a tiny targeted set is an encouraging regression signal, not broad generalization. Citation coverage is low, and the smoke set does not compute factual faithfulness.

02 / PRODUCT FRAMING

Problem definition and success criteria

Problem

Agricultural research is distributed across long PDFs, tables, terminology variants, and study-specific conditions. A model prompted directly from a broad question can produce plausible but unsupported advice, combine findings from different crops, or lose the referent of a short follow-up. SAWVIAL separates evidence preparation, retrieval, answer generation, and response checks.

Engineering objectives

- Evidence traceability: preserve source document, page/section, knowledge-unit ID, and source text through retrieval and response.
- Domain discipline: constrain supported subject areas and return explicit scope/no-evidence responses when evidence is absent or weak.
- Study-condition fidelity: preserve crop, soil, location, season, units, and numeric conditions; never transfer irrigation details across crops.
- Conversation continuity: use recent relevant questions and answers to resolve references and formatting instructions.
- Offline reproducibility: keep canonical JSONL as the source of truth and make derived indexes rebuildable.
- Observable quality: evaluate retrieval, citation behavior, follow-up resolution, formatting, and latency.

System boundary

This is a retrieval-augmented research interface, not a field-level agronomy decision-support system. Corpus coverage is not guaranteed for every crop, region, climate, cultivar, soil, or growth stage. A generated response is not a substitute for local extension guidance, replication, or expert agronomic advice.

Safe interpretation

Report what the source says, distinguish a study result from a source-stated recommendation, and name missing details rather than filling them with generic assumptions.

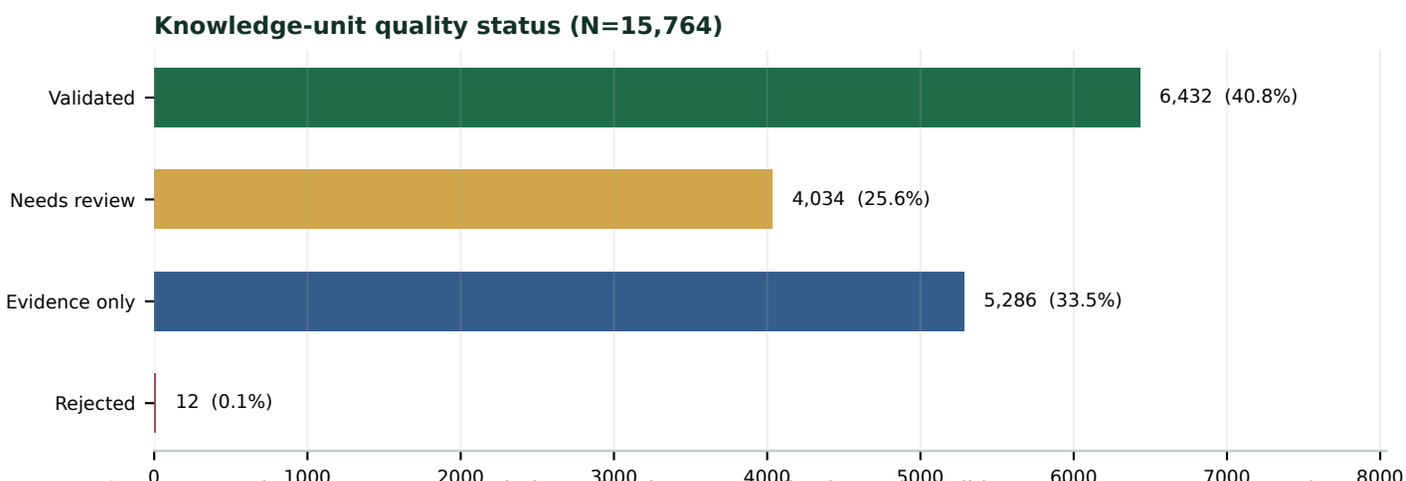
03 / CORPUS INVENTORY

Dataset composition and semantics

Corpus size and recorded coverage

Dataset asset	Count / property	Role
documents.jsonl	186 documents	Deduplicated document metadata and build counts.
knowledge_units.jsonl	15,764 units	Canonical evidence, source, context, classification, QA, and quality fields.
qa.jsonl	6,563 Q&A pairs	Generated question-answer records linked to knowledge-unit IDs.
Index vectors	15,752 x 768	Derived retrieval representation with an explicit vector-position-to-ID map.
Average evidence length	620.4 characters	Mean evidence string length in stored statistics.
Input corpus	186 research papers	PDFs in the project paper collection.

Quality states: units remain auditable

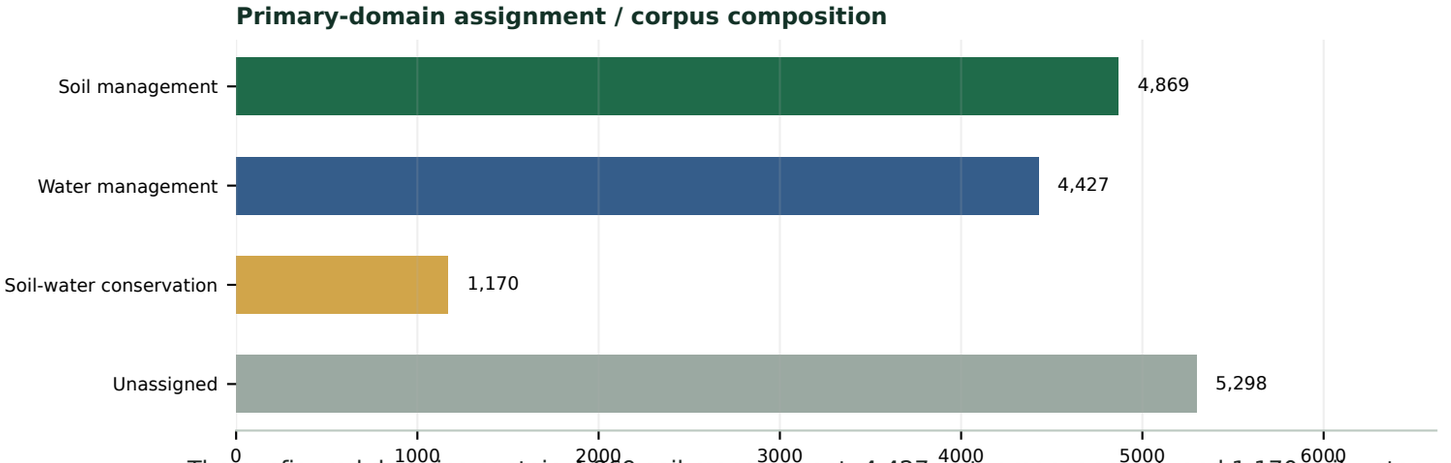


Statuses are pipeline outcomes, not independent human certifications. The builder retains status and reason data rather than silently discarding low-confidence material. At answer time, the RAG passage builder admits validated and needs_review records and then applies usability/relevance filters; evidence_only and rejected records are not answer evidence.

04 / DATA MODEL

Taxonomy and knowledge-unit schema

Domain balance



The configured domains contain 4,869 soil-management, 4,427 water-management, and 1,170 soil-water-conservation units. 5,298 units are unassigned/evidence-only. This imbalance is important: counts describe corpus prevalence, not research importance.

Record anatomy

Field family	Meaning in the RAG pipeline
id + source	Stable SWM ID; title/authors/year where extracted; type, page, section.
classification	Primary/secondary domain, subdomains, topics, classifier confidence.
context	Crop, variety, growth stage, location, climate zone, season, soil, irrigation method.
knowledge	Evidence; extracted finding; optional explicit recommendation and recommendation type.
structured_facts	Raw and parsed numbers/units/ranges/effects when recognized.
qa + retrieval	Evidence-derived Q&A and semantic_text/keywords representation.
quality + provenance	Validation status/reasons and document/page/section/source hashes.

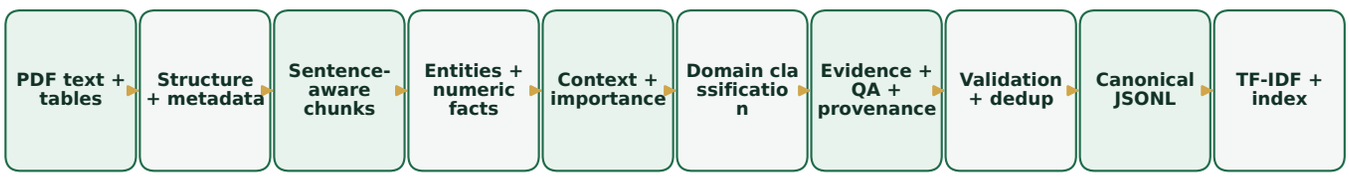
Why atomic records

Retrieving a study result at the granularity where crop, treatment, and measurement stay together makes evidence selection and citation auditing safer than flattening a PDF.

05 / IMPLEMENTATION PIPELINE

Document processing and data preparation

End-to-end transform



Dataset-first principle: JSONL is the canonical, inspectable source of truth. Embeddings and indexes are derived products and can be rebuilt.

Preparation decisions

- Build metadata records software 2.0.0, taxonomy 1.0.0, Python 3.10.10, and chunk bounds 80-1600 characters (target 700).
- PDF extraction is page-aware and table-aware; document structure is converted to sections/paragraphs before chunking.
- Concept aliases normalize surface forms; crop, soil, season, location, and irrigation method are attached as context.
- Numeric extraction stores raw forms alongside parsed measurements. Recommendations are separated from findings and only captured when explicit language exists.
- Deduplication precedes stable sequential IDs. Validation records a status and reason for each unit.

Quality-control caveat

Validation checks evidence length, provenance, allowed taxonomy domains, importance/confidence thresholds, and a crude disconnected-QA condition. It is useful pipeline QC, but not expert review of scientific validity, OCR accuracy, extraction completeness, or causal interpretation.

06 / STANDARD ALGORITHM IDENTIFICATION

Retrieval algorithm and indexing

Configured index artifact

Persisted index metadata identifies IndexFlatIP (exact cosine, vectors L2-normalized) with 15,752 vectors of dimension 768 using tfidf in offline mode. An explicit ID map preserves vector-to-knowledge-unit identity.

Runtime offline retrieval

For the TF-IDF backend, query-time code builds/caches a scikit-learn sparse TF-IDF matrix over corpus semantic_text, normalizes English terms, uses word unigrams and bigrams, then ranks sparse matrix-query dot products. The default vectorizer L2-normalizes rows, making the dot product cosine similarity for normalized vectors. This runtime branch does not call FAISS search; FAISS is persisted as an index artifact and used by the alternative dense backend path. This distinction is important for an accurate implementation description.

Relevance pipeline

Question normalization and narrow query expansion -> top candidate ranking -> domain/crop/context filtering -> quality/usability checks -> crop-labeled evidence narrowing -> top-k passages -> minimum-score gate.

Score threshold and trade-off

The API uses a minimum retrieval score of 0.11 for the no-evidence gate. A fixed threshold is simple but score distributions depend on wording and query expansion. Calibrate it on held-out in-domain and out-of-domain queries. Retrieval rank alone does not prove that a passage supports every generated claim.

Method basis: TF-IDF term weighting (Salton and Buckley, 1988); vector similarity and ranked retrieval evaluation (Manning, Raghavan, and Schutze, 2008).

07 / RAG EXECUTION

Answer generation and conversation handling



The API serializes generation with a process lock, bounds context to at most 24 turns and 8,000 characters, supports token-generation cancellation, and returns source metadata.

Model and prompt contract

Default answer model: Qwen/Qwen2.5-1.5B-Instruct, loaded from the local Hugging Face cache. The LoRA adapter is disabled by default. The system prompt requests relevant-record-only claims, explicit missing-detail statements, exact conditions/units, crop-specific interpretation, and [SWM_...] citations. With translation enabled, non-English turns are translated into English and the answer is translated back.

Potato evidence-isolation example

SWM_000671 contains the crop-labeled clause "potato: 3 irrigations applied" alongside information about other crops. The filter isolates the potato clause and removes unrelated crop findings/facts. The bounded response is: "The dataset records 3 irrigations for potato; it does not specify the timing or water amount. [SWM_000671]" This is not a full potato irrigation schedule.

Follow-up resolution

The browser sends recent messages. The API searches backward for the most recent substantive in-domain question, skipping formatting-only and follow-up turns. A request such as "write it in one line" is combined with the active topic and recent conversation before retrieval and prompting. The regression suite includes water-conservation follow-ups.

08 / DEMONSTRATED ENGINEERING

Tools and implementation methods

Layer	Tools / method	Purpose
Corpus	Python 3.10; PyMuPDF; JSONL	Local extraction and inspectable interchange.
Retrieval	scikit-learn TF-IDF; sparse product; FAISS artifact	Offline weighted retrieval; no hosted embedding API.
Generation	PyTorch + Transformers; Qwen2.5-1.5B-Instruct	Local instruction model with retrieved evidence.
API	FastAPI + Pydantic	Typed request boundary, validation, cancellation.
UI	HTML/CSS/JavaScript; fetch; AbortController	Browser chat, recent context, request/audio controls.
Evaluation	evaluate_rag.py; JSON report	Repeatable smoke and dataset-QA runs.
Optional speech /translation	MMS-VITS; NLLB-200	Product features; not scored in answer-quality smoke tests.

Engineering practices visible in the implementation

- Separation of concerns across ingestion, chunking, extraction, taxonomy, writing, indexing, retrieval, API, UI, and evaluation.
- Traceability from stable knowledge-unit IDs to evidence and PDF provenance.
- Bounded question length, history size, top-k, cancellation IDs, and speech text length.
- Explicit scope/no-evidence responses and surfaced API/model failures rather than success-shaped fallbacks.
- Build metadata records software/taxonomy/Python versions and extraction/chunk thresholds.

Active vs available

The saved index metadata says TF-IDF/offline; the default LoRA adapter is disabled; speech and translation are outside the answer-quality smoke metrics.

09 / METRIC DEFINITIONS

Standard evaluation algorithms and metrics

Retrieval metrics implemented

Metric	Definition / interpretation	Evaluator
Precision@k	|top-k retrieved intersect relevant| / k; fixed k denominator in this implementation.	Implemented; k defaults to 5.
Recall@k	Relevant IDs found in top k / all relevant IDs; evidence coverage.	Implemented when gold IDs exist.
MRR	Mean reciprocal rank of first relevant result, 1/rank.	Per-case reciprocal rank, then mean.
nDCG@k	Discounted gain normalized by ideal ordering; binary ID relevance here.	Implemented with log2 rank discount.

Answer and system metrics

Metric	Definition / limitation
Token F1	Harmonic mean of token-overlap precision and recall vs. reference; lexical, not semantic.
ROUGE-L F1	Longest-common-subsequence overlap; paraphrase-sensitive; only with reference_answer.
Grounded rate	Fraction whose API grounded flag is true; not independent factual verification.
Cited-source ratio	Per-case cited returned sources / all returned sources; averaged across cases.
One-line compliance	No newline when one_line is expected; mean over formatting cases only.
Latency	Evaluator wall-clock request time in ms; affected by prompt, hardware, warm-up, load.

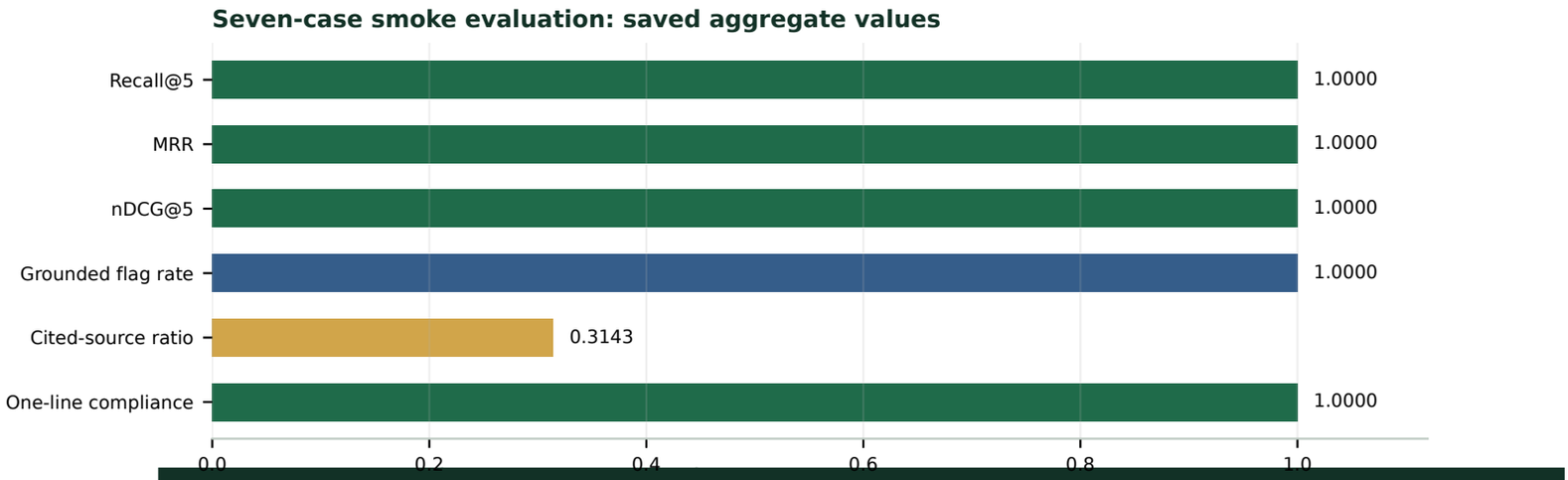
Not computed in this evaluator: BLEU, BERTScore, answer relevancy, context precision/recall, calibration, abstention accuracy, or a human/LLM faithfulness score. Text-overlap scores cannot replace factual support assessment.

10 / MEASURED EVIDENCE

Evaluation protocol and preliminary results

Saved seven-case live-API snapshot

Values below are read from the project-root evaluation-results.json. The evaluator does not currently write a generated_at timestamp, so these values are the saved report snapshot, not a fully time-stamped experiment manifest.



Metric	Value	Qualification
Cases	7	Targeted regression prompts; not a random user sample.
Precision@5	0.3143	Fixed k=5; smoke gold sets contain one to three IDs.
Recall@5	1.0000	All listed gold IDs found.
MRR	1.0000	First relevant ID ranked first in these cases.
nDCG@5	1.0000	Binary relevance labels; perfect for this set.
Grounded flag	1.0000	API flag, not a factuality judge.
Cited-source ratio	0.3143	Citation coverage gap.
One-line compliance	1.0000	Two formatting tests.
Mean latency	26,730.8 ms	Not a production service target.

Answer similarity was not measured

token_f1 and rouge_l_f1 are null because smoke cases have no reference_answer. Null means not measured; it does not mean zero or perfect.

11 / WHAT THE RUN DEMONSTRATES

Case-level verification and findings

Test case	Observed	Returned evidence / checks
farm-water-conservation	grounded=True; latency=74592 ms	IDs=SWM_012405,SWM_012799,SWM_012408; citation=0.00
contextual-water-benefit	grounded=True; latency=26390 ms	IDs=SWM_012405,SWM_012408,SWM_012799; citation=0.00
water-one-line-after-followups	grounded=True; latency=7818 ms	IDs=SWM_012405,SWM_012799,SWM_012408; citation=0.20; one-line=1.0
potato-irrigation	grounded=True; latency=2118 ms	IDs=SWM_000671; citation=1.00
potato-one-line-followup	grounded=True; latency=2118 ms	IDs=SWM_000671; citation=1.00; one-line=1.0
soil-fertility	grounded=True; latency=42898 ms	IDs=SWM_003637,SWM_002322,SWM_002555; citation=0.00
irrigation-efficiency	grounded=True; latency=31182 ms	IDs=SWM_012781,SWM_014375,SWM_012778; citation=0.00

What can be concluded

- The water-conservation and context follow-up cases retrieved their expected SWM_012405 record at rank 1 in this snapshot.
- The potato path returns a narrow evidence-backed result without borrowing adjacent-crop irrigation methods.
- Citation compliance for general answers is unreliable: the prompt requests citations, but the API currently records uncited generations rather than rejecting/retrying them.
- Seven curated cases are a regression sanity check, not representative evaluation of 15,764 records or real-user query distributions.

12 / TERMINAL WORKFLOW

Run instructions and reproducibility

1. Start the API in Terminal A

PowerShell

```
cd "C:\Users\KUNAL DEBNATH\Desktop\sawvial_project" cd "C:\Users\KUNAL DEBNATH\Desktop\sawvial_project" python api.py
```

Keep this terminal running. Default URL: http://localhost:8000.

2. Run the seven-case smoke suite in Terminal B

PowerShell

```
cd "C:\Users\KUNAL DEBNATH\Desktop\sawvial_project" python evaluate_rag.py --smoke --output evaluation-results.json
```

This prints per-case progress and aggregate metrics, then writes detailed JSON. The seven live model calls may take minutes; request latency can be tens of seconds. The named output file is overwritten.

3. Evaluate a larger dataset-QA sample

PowerShell

```
cd "C:\Users\KUNAL DEBNATH\Desktop\sawvial_project" python evaluate_rag.py --limit 25 --top-k 5 --seed 42 --output evaluation-results-25.json
```

This samples up to 25 eligible dataset/qa.jsonl rows using a deterministic shuffle seed and adds token F1 / ROUGE-L. These Q&A pairs are corpus-derived, not independent human gold annotations; interpret lexical similarity cautiously.

Additional commands

- python build_dataset.py --output_dir ./dataset --stats
- python build_dataset.py --output_dir ./dataset --index_dir ./index --query "potato irrigation" --top_k 5
- Get-Content .\evaluation-results.json (PowerShell: inspect detailed JSON)

Repeatability controls

Dataset build metadata records Python/software/taxonomy versions and chunk/extraction thresholds. Evaluator seed controls sample selection only. Model generation is sampled (do_sample=True, temperature=0.35, top_p=0.9) without a fixed model RNG seed, so wording and latency vary. Record run time, code/data/index hashes, model/hardware versions, warm/cold state, and generation seed in a future manifest.

13 / NEXT STEPS

Limitations and prioritized improvement plan

Observed limitations

- Citation coverage: saved mean cited-source ratio is 0.3143; multiple answers contain no inline citation.
- No independent answer-faithfulness judge: grounded_rate is an API state flag, not claim entailment.
- Benchmark validity: smoke cases are curated; corpus Q&A is evidence-derived, not held-out expert annotation.
- Domain labels are imbalanced; 5,298 units have no primary domain, and conservation has fewer units than soil or water management.
- The fixed 0.11 threshold and hand-built query expansions may not generalize to unseen phrasing/languages.
- The report has no timestamp/model hash/generation seed; latency varies with device, warm-up, and prompt size.

Prioritized plan

Priority	Action	Acceptance evidence
P0	Validate citations against retrieved IDs; retry once or abstain when a factual answer lacks citations.	No invalid IDs; claim-level citation audit.
P0	Build a held-out expert-reviewed benchmark with gold IDs, answer facts, OOD labels, and context turns.	Annotation guide, reviewed subset, split by source paper.
P1	Add claim-level support/completeness review; calibrate a judge only after human agreement checks.	Supported/unsupported labels and reviewer agreement.
P1	Tune retrieval threshold on in-domain, hard-negative, and OOD queries.	Precision/recall and abstention curves; untouched final test split.
P2	Add run manifest, fixed generation seed, repeated runs, and p50/p95 latency.	Timestamp, revision, hashes, models, hardware, generation settings.
P2	Audit extraction/classification and test language-specific query coverage.	Stratified error and coverage tables.

Recommended acceptance framework

Do not optimize a single scalar. Require held-out retrieval quality, citation correctness, claim support, follow-up resolution, OOD abstention, and p50/p95 latency.

14 / TECHNICAL NOTES

Standard algorithms: theory to implementation

TF-IDF retrieval

TF-IDF weights a term by its within-document frequency and inverse corpus frequency, reducing the influence of ubiquitous words and emphasizing discriminative terms. SAWVIAL uses sublinear term frequency, stop-word removal, custom plural normalization, and word n-grams of orders 1 and 2. L2-normalized query/document vectors are ranked by dot product, equivalent to cosine similarity for normalized vectors. This is interpretable and offline, but lexical overlap limits synonym and cross-language recall.

Top-k ranked evaluation

For query q, gold relevant IDs R(q), and ranked results r_1...r_k, precision measures relevance density; recall measures coverage. Reciprocal rank 1/rank rewards the first relevant result near the top. DCG discounts lower ranks by log2(rank+1); nDCG normalizes by ideal DCG. This evaluator uses binary ID membership, not graded human relevance.

Retrieval-augmented generation

RAG separates parametric model knowledge from non-parametric evidence retrieved at inference. SAWVIAL retrieves records, places IDs/findings/context/evidence into the prompt, then asks a local decoder-only instruction model to synthesize an answer. Retrieval reduces dependence on memorized corpus facts but does not guarantee faithfulness; retrieval recall, evidence quality, citation checks, and abstention remain necessary.

Complementary measures are not substitutes

Token F1 and ROUGE-L estimate surface overlap. BERTScore uses contextual representations for semantic similarity but can still reward a close yet factually incorrect answer. Human or calibrated claim-level evaluation is needed for factual support. Citation presence is weaker than citation correctness: the cited passage must support the adjacent claim.

15 / SOURCES

References and report provenance

Method references

[1] Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. Information Processing & Management, 24(5), 513-523. https://doi.org/10.1016/0306-4573(88)90021-0

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[3] Jarvelin, K., & Kekalainen, J. (2002). Cumulated gain-based evaluation of IR techniques. ACM Transactions on Information Systems, 20(4), 422-446. https://doi.org/10.1145/582415.582418

[4] Lin, C.-Y. (2004). ROUGE: A Package for Automatic Evaluation of Summaries. Text Summarization Branches Out, ACL, 74-81. https://aclanthology.org/W04-1013/

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[6] Johnson, J., Douze, M., & Jegou, H. (2019). Billion-scale similarity search with GPUs. IEEE Transactions on Big Data, 7(3), 535-547. https://doi.org/10.1109/TBDATA.2019.2921572

[7] FAISS documentation: MetricType and distances. https://github.com/facebookresearch/faiss/wiki/MetricType-and-distances

[8] scikit-learn documentation: Feature extraction of text data; TfidfVectorizer. https://scikit-learn.org/stable/modules/feature_extraction.html#text-feature-extraction

Project artifacts consulted

README.md; api.py; evaluate_rag.py; llm/common.py; llm/rag_chat.py; sawvial/pipeline.py; sawvial/config.py; sawvial/validation/validator.py; sawvial/embeddings/embedder.py; sawvial/index/retriever.py; sawvial/index/faiss_builder.py; dataset/dataset_statistics.json; dataset/build_metadata.json; index/index_metadata.json; evaluation-results.json.

Provenance and limits

Numeric corpus and evaluation values were read from local project artifacts. The saved evaluation JSON contains seven cases but no generated_at field. References identify standard methods; they do not independently validate this project's performance. This report describes the current implementation and preliminary results, not an independent scientific audit.

Presentation takeaway

SAWVIAL has a traceable corpus-to-retrieval-to-chat implementation and a first regression evaluator. The next critical step is a human-reviewed held-out benchmark with reliable claim-level citations and a measured faithfulness/abstention protocol.